GROOMER AVAILABILITY – API ISSUES & DEFECTS REPORT

Complete Backend API Testing, Negative Testing, Conflict Handling & Availability Engine Audit

TARGET ENVIRONMENT	STORE & STYLIST	TOTAL API DEFECTS FOUND	TEST DATE
PCMDEV Production / Staging	Darwin (SHEAR-001) / Richard Cooke (ID: 2)	8 Verified Backend Issues	November 04, 2026

EXECUTIVE SUMMARY & API HEALTH STATUS: Thorough end-to-end API testing across all 9 availability phases revealed **8 critical and high-severity backend defects**. Most critically, the booking creation API (`POST /api/bookings`) completely bypasses groomer break and leave availability checks, allowing customers to book appointments directly inside scheduled breaks and approved time-off. Furthermore, the availability calculation engine (`GET /api/availability`) contains an internal architectural discrepancy where `availableSlots` calculates open hours while `timeSlotChoices` marks all afternoon slots unavailable.

API BUG 01 — Booking API Creates Appointments During Scheduled Groomer Break

CRITICAL

Endpoint: `POST /api/bookings` **Category:** Business Logic & Conflict Validation **Date Tested:** 2026-11-19 **Affected Entity:** Groomer Richard Cooke (ID: 2)

EXPECTED RESULT	ACTUAL RESULT
The API must validate requested appointment times against active groomer breaks (configured: 12:00 - 13:00, Break ID: 14) and reject the request with **HTTP 409 Conflict** or **HTTP 400 Bad Request**: `{"success": false, "message": "Groomer is on a scheduled break at this time"}`	The API returned **HTTP 201 Created** and successfully inserted **Booking ID 150** directly into Richard Cooke's break period (12:00 - 13:15). The break schedule was completely ignored.

HTTP POST REQUEST PAYLOAD (BREAK OVERLAP):

```
POST https://api.shearheavenpetspa.com/api/bookings
{
  "storeId": "SHEAR-001", "clientId": "SHEAR-001", "regionId": "DWG-001",
  "petId": 4, "serviceId": 193, "groomerId": 2,
  "bookingDate": "2026-11-19", "startTime": "12:00", "endTime": "13:15",
  "notes": "Negative Test - Break Overlap Attempt"
}
```

DEFECTIVE HTTP 201 RESPONSE:

```
{
  "status": 201,
  "body": {
    "success": true,
    "message": "Booking created successfully",
    "data": { "bookingId": 150, "status": "pending", "serviceStage": "pending", "totalPrice": 85, "finalTotal": 91.91 }
  }
}
```

API BUG 02 Booking API Creates Appointments During Approved Time-Off / Leave

CRITICAL

Endpoint: POST /api/bookings

Category: Business Logic & Conflict Validation

Date Tested: 2026-11-18

Affected Entity: Groomer Richard Cooke (ID: 2)

EXPECTED RESULT	ACTUAL RESULT
The API must verify approved groomer time-off records (configured: 2026-11-18 10:00 - 12:00, Leave ID: 13) and reject conflicting bookings with **HTTP 409 Conflict**: `{"success": false, "message": "Groomer is on approved leave during this time"}`	The API returned **HTTP 201 Created** and successfully inserted **Booking ID 151** for 10:00 - 11:15 on 2026-11-18, directly overlapping the groomer's medical leave window.

HTTP POST REQUEST PAYLOAD (LEAVE OVERLAP):

```
POST https://api.shearheavenpetspa.com/api/bookings
{
  "storeId": "SHEAR-001", "clientId": "SHEAR-001", "regionId": "DWG-001",
  "petId": 4, "serviceId": 193, "groomerId": 2,
  "bookingDate": "2026-11-18", "startTime": "10:00", "endTime": "11:15",
  "notes": "Negative Test - Time Off Overlap Attempt"
}
```

DEFECTIVE HTTP 201 RESPONSE:

```
{
  "status": 201,
  "body": {
    "success": true,
    "message": "Booking created successfully",
    "data": { "bookingId": 151, "status": "pending", "serviceStage": "pending", "totalPrice": 85, "finalTotal": 91.91 }
  }
}
```

API BUG 03 Internal Discrepancy Between availableSlots and timeSlotChoices in Availability Engine

HIGH

Endpoint: GET /api/availability

Category: Slot Computation & Payload Consistency

Date Tested: 2026-11-18

Impact: Blocks 100% of afternoon bookings

EXPECTED RESULT	ACTUAL RESULT
For Wednesday 2026-11-18 (working hours 10:00 - 15:00, leave 10:00 - 12:00), both `availableSlots` and `timeSlotChoices` must agree. Slots after 12:00 PM (12:00 - 13:15, 13:15 - 14:30) must be returned with `isAvailable: true` and `status: "available"`.	In the **exact same response payload**: • `availableSlots` lists **2 available slots** (12:00-13:15, 13:15-14:30). • `timeSlotChoices` lists **4 slots, ALL marked isAvailable: false** (10:00, 11:15, 12:30, 13:45). Because frontend renders timeSlotChoices, no booking can occur!

API RESPONSE DISCREPANCY EVIDENCE (GET /API/AVAILABILITY?DATE=2026-11-18...):

```
// 1. availableSlots says afternoon IS AVAILABLE:
"availableSlots": [
  { "startTime": "12:00", "endTime": "13:15", "groomerId": 2, "isAvailable": true },
  { "startTime": "13:15", "endTime": "14:30", "groomerId": 2, "isAvailable": true }
]

// 2. BUT timeSlotChoices in the same response says afternoon IS UNAVAILABLE:
"timeSlotChoices": [
  { "startTime": "10:00", "endTime": "11:15", "status": "unavailable", "isAvailable": false },
  { "startTime": "11:15", "endTime": "12:30", "status": "unavailable", "isAvailable": false },
  { "startTime": "12:30", "endTime": "13:45", "status": "unavailable", "isAvailable": false },  // BUG
  { "startTime": "13:45", "endTime": "15:00", "status": "unavailable", "isAvailable": false }   // BUG
]
```

API BUG 04 Rigid Interval Slicing Drops Valid Post-Break Slots in timeSlotChoices

HIGH

Endpoint: GET /api/availability

Category: Interval Calculation & Alignment

Date Tested: 2026-11-19

Impact: Afternoon slots omitted from client options

EXPECTED RESULT	ACTUAL RESULT
When a break is configured (12:00 - 13:00), the slot generator should realign available slots starting from 13:00 (13:00 - 14:15, 14:15 - 15:30) across both `availableSlots` and `timeSlotChoices`.	`availableSlots` generated 13:00 - 14:15 and 14:15 - 15:30, but `timeSlotChoices` rigidly stepped from 09:00 (09:00, 10:15, 11:30, 12:45, 14:00). Slot 13:00 - 14:15 was omitted, and 14:00 - 15:15 was marked `status: "unavailable", isAvailable: false`.

API RESPONSE EVIDENCE (GET /API/AVAILABILITY?DATE=2026-11-19...):

```
"availableSlots": [
  { "startTime": "09:00", "endTime": "10:15", "isAvailable": true },
  { "startTime": "10:15", "endTime": "11:30", "isAvailable": true },
  { "startTime": "13:00", "endTime": "14:15", "isAvailable": true },
  { "startTime": "14:15", "endTime": "15:30", "isAvailable": true }
]
// timeSlotChoices dropped 13:00 - 14:15 and marked 14:00 - 15:15 unavailable:
"timeSlotChoices": [
  { "startTime": "11:30", "endTime": "12:45", "status": "break", "isAvailable": false },
  { "startTime": "12:45", "endTime": "14:00", "status": "break", "isAvailable": false },
  { "startTime": "14:00", "endTime": "15:15", "status": "unavailable", "isAvailable": false }
]
```

API BUG 05 Inverted Working Hours Accepted Without Chronological Validation

HIGH

Endpoint: PUT `/api/common/groomer/availability/working-hours/:id` **Category:** Input Validation Failure **Tested Values:** startTime: 17:00, endTime: 09:00

EXPECTED RESULT	ACTUAL RESULT
The API should validate that `endTime > startTime` and reject inverted time ranges with **HTTP 400 Bad Request**: `{"success": false, "message": "End time must be after start time"}`	The API returned **HTTP 200 OK** and saved `startTime: "17:00", endTime: "09:00"` directly to the database, creating a corrupt inverted schedule.

DEFECTIVE RESPONSE:

```
PUT /api/common/groomer/availability/working-hours/12
Body: {"isWorking":true,"startTime":"17:00","endTime":"09:00"}
Response: 200 OK
{
  "success": true, "message": "Working day updated",
  "data": { "id": 12, "dayOfWeek": "Thursday", "isWorking": true, "startTime": "17:00", "endTime": "09:00" }
}
```

API BUG 06 Inverted Time-Off / Break Duration Accepted Without Validation

HIGH

Endpoint: POST `/api/common/groomer/availability/time-off` **Category:** Input Validation Failure **Tested Values:** startTime: 16:00, endTime: 11:00

EXPECTED RESULT	ACTUAL RESULT
The API should validate time boundaries and reject requests where `endTime <= startTime` with **HTTP 400 Bad Request**.	The API returned **HTTP 200 OK** and generated **Time-Off ID 15** with `startTime: "16:00", endTime: "11:00"`.

DEFECTIVE RESPONSE:

```
POST /api/common/groomer/availability/time-off
Body: {"leaveType":"break","reason":"Negative Test - Inverted Break","startDate":"2026-11-27","endDate":"2026-11-27","startTime":"16:00","endTime":"11:00"}
Response: 200 OK
{
  "success": true, "message": "Time off saved",
  "data": { "id": 15, "startTime": "16:00", "endTime": "11:00", "leaveType": "break" }
}
```

API BUG 07 Overlapping Time-Off / Leave Entries Allowed for Same Groomer

MEDIUM

Endpoint: POST `/api/common/groomer/availability/time-off` **Category:** Conflict & Collision Handling **Tested Values:** 10:00-12:00 vs 11:00-13:00 on 2026-11-28

EXPECTED RESULT	ACTUAL RESULT
The API should check for existing time-off entries for the groomer and reject overlapping submissions with **HTTP 409 Conflict**.	The API returned **HTTP 200 OK** and created duplicate overlapping records (ID: 16 and ID: 17) for the same groomer.

API BUG 08 **Zero-Duration Time-Off / Break Allowed (startTime == endTime)** MEDIUM

Endpoint: POST `/api/common/groomer/availability/time-off`

Category: Boundary Validation Failure

Tested Values: startTime: 12:00, endTime: 12:00

EXPECTED RESULT	ACTUAL RESULT
A non-zero time range must be enforced, rejecting identical start and end times with **HTTP 400 Bad Request**.	The API returned **HTTP 200 OK** and created **Time-Off ID 18** with 0 minutes duration (12:00 - 12:00).